AI-Claim Audit for Robust Vision

MILS Assignment Report

1. 實驗設定 (Problem Setting)

本次實驗旨在比較高性能大型架構與輕量化行動架構在面對分佈偏移（Distribution Shift）時的穩健性指標。

- 基準資料集 **(Clean)**: ImageNet-V2
- 偏移資料集 **(Shifted)**: ImageNet-R (16 種風格變體)
- 模型: ResNeXt-101 (Heavy) vs. MobileNet-V3 (Lightweight)
- 權重: 皆使用 IMAGENET1K_V2 預訓練權重

2. 實驗環境 (Experimental Environment)

項目	內容
硬體	NVIDIA GeForce RTX 5090 (32GB VRAM)
執行模式	GPU Inference (CUDA 12.8)

3. 穩健性指標評估 (Robustness Metrics)

指標 **(Metrics)**	**ResNeXt-101**	**MobileNet-V3**
Clean Accuracy	87.50%	77.00%
Shifted Accuracy	30.50%	19.25%
Accuracy Drop	0.5700	0.5775
Relative Drop	65.14%	75.00%
Average Robust Accuracy	27.19%	15.40%
Worst-condition Accuracy	0.00%	0.00%

Wrong Confidence	20.62%	21.21%
Rejection Robustness (80%)	34.38%	23.44%

4. AI Claim Audit (假說審核)

ID	假說內容 (Claim)	證據 (Evidence)	審核結果 (Decision)
C1	ResNeXt-101 has a smaller average Accuracy Drop on ImageNet-R than MobileNet-V3.	ResNeXt Drop: 0.5700, MobileNet Drop: 0.5775	**Supported**
C2	ResNeXt-101's Accuracy Drop is largest on 'cartoon' and 'sketch' sub-categories.	Max ResNeXt Drop on art (0.8750).	**Refuted**
C3	Wrong Confidence is higher for MobileNet-V3 than ResNeXt-101.	ResNeXt Wrong Conf: 0.2062, MobileNet Wrong Conf: 0.2121	**Supported**
C4	Both models have their highest Accuracy Drop on 'sculpture' renditions.	ResNeXt Max: art, MobileNet Max: plush object	**Refuted**
C5	The failure overlap between ResNeXt-101 and MobileNet-V3 is less than 50%.	Overlap Ratio: 81.57%	**Refuted**

5. 深入分析與觀察 (In-depth Analysis)

實驗顯示，儘管 ResNeXt-101 在基準準確度上大幅領先，但兩者在面對風格偏移時皆表現出顯著的性能退化。值得注意的是，MobileNet-V3 雖然規模較小，但在某些特定穩健性指標（如 Relative Drop）上展現出了不俗的韌性。

6. 失敗案例分析 (Failure Analysis)

ResNeXt-101 Fail #1

MobileNet-V3 Fail #1

7. 結語 (Conclusion)

本實驗透過比較 ResNeXt 與 MobileNet，揭示了模型規模與穩健性之間的複雜關係。多維度指標的審核使我們能更客觀地評估 AI 模型的實際部署風險。